The historical CH density instrument

Controls: Historical CH instrument, construction method and coverage; MSA and State.

1. Describe historical settlement within a modern region. Let g be an MSA or state, P_j historical population and a_j historical mapped area in square kilometres. The historical year is the selected census year. Reporting areas j are counties, or combined counties when population was reported jointly. Let s_{jg} be the fraction of area j inside g. Population is assigned by geographic overlap. Define log density $\ell_j = \log(P_j/a_j)$ and population weights $w_{jg} = s_{jg}P_j / \sum_k s_{kg}P_k$. The historical CH factor is

$$H_g(\theta_0) = \sum_j w_{jg} \exp\big((\theta_0 - 1)\ell_j\big).$$

Only known positive populations enter these weights. Geographic splits change the assigned population, while preserving the original source density.

2. Derivative instrument. The first method uses the slope of $\log H_g$ at $\theta_0 = 1$:

$$z_g = \left.\frac{\partial \log H_g(\theta_0)}{\partial \theta_0}\right|_{\theta_0=1} = \sum_j w_{jg}\ell_j.$$

This describes log density around an average historical resident. Although $H_g(1) = 1$ everywhere, its slope differs across regions. No preliminary elasticity estimate is needed. This instrument stays fixed while GMM estimates the modern productivity elasticity $\rho = \theta - 1$.

3. NLS-based instrument. First estimate the modern CH equation by nonlinear least squares (NLS), using the same outcome, regional sample, controls and county adjustments as the final model. Call this estimate $\widehat{\theta}_{\text{NLS}}$. Construct

$$z_g = \frac{\log H_g(\widehat{\theta}_{\text{NLS}})}{\widehat{\theta}_{\text{NLS}} - 1}.$$

The denominator only rescales the instrument; at $\widehat{\theta}_{\text{NLS}} = 1$, use the derivative limit. GMM holds the historical instrument fixed while estimating structural θ in the full modern CH model. With one instrument and one elasticity, one-step and two-step GMM give the same solution. Inference conditions on the NLS estimate. Since NLS uses productivity, the usual fixed-instrument Stock–Wright test is not reported.

4. Choose the coverage requirement. Choose *80%*, *95%*, or *No coverage filter*. Coverage is known-population area divided by full mapped area in both years. Recorded zeros count; unknown populations do not. No filter still requires positive population and a finite instrument. Under *States Only*, State observations require admission by the end of the territory year; State sources use year-end status in each census year. Coverage cannot override admission eligibility. Area-weighted population retains its separate 95% rule. Neither CH method establishes instrument validity.